SRN: PES1UG24AM462

NAME: GREESHMA G
AIML SEC-H
SE LAB 3

**LAB 3: COMPONENT MODELLING & ARCHITECTURAL PATTERN SELECTION**

**Problem Statement 37: Group Travel Expense Simplification Engine**

**Architecture Selection**

**We chose Microservices Architecture for the Group Travel Expense Simplification Engine.**

**Architectural Choice**

The Group Travel Expense Simplification Engine is a multi-currency expense-sharing system that records group expenses, converts currencies using daily exchange rates, calculates traveler balances, and determines minimum-transaction debt settlements. The system is divided into separate components such as Expense Management, Currency Conversion, Data Management, and Settlement Calculation.

**Reason 1 – Independent Components**

Microservices architecture allows the major functions of the system to operate as separate services. For example, Currency Conversion and Settlement Calculation can work independently. This makes the system modular and allows individual components to be modified or maintained without affecting the entire application.

**Reason 2 – Scalability and Fault Isolation**

Different functions may have different processing requirements. Settlement Calculation may require more processing when there are many travelers and expenses, while other services may have lower workloads. Microservices allow individual services to be scaled independently. A failure in one service can also be isolated so that it does not necessarily bring down the complete system.

**Security Advantage**

Each service can have its own authentication and authorization controls, limiting access to sensitive expense and traveler information. Communication between services can also be secured using authenticated API requests and encryption.

**Performance Benefit**

Microservices allow computationally intensive operations such as settlement calculation and currency conversion to be handled independently. These services can be scaled separately when the workload increases, helping the system maintain good performance for larger groups and more expenses.

UML DIAGRAM:

